# Yang–Mills Branch Book

# Spectral-Coercive Realization, Gap-Subcritical Persistence, and the Closure Ledger

Elements of Nested Fibrational Cosmology

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## Front Block

The following five items constitute the mandatory intake declaration required by Book VII (Canonical Closure Schema Theorem, Thm. VII.6.4) and Book V (UBLT, Thm. V.7.4; Certified Branch Legitimacy Theorem, Thm. V.8.1).

Imported spine results.

(1) Book I: Observational Quotient Theorem (Thm. I.6.1); $K_0$ primality (Thm. I.3.7); Structural Entropy and Entropy Monotonicity (Def. I.5.1, Thm. I.5.2); Collapse Gates CG-1–CG-6 (SRs I.1.4–I.1.9).
(2) Book II: Boundary Stabilization Theorem (Thm. II.8.1); Phase-A/B/C structural classification (Def. II.5.1); PASS as Derived Necessity (Thm. II.5.14); Binary Rigidity (Thm. II.5.8); Bit-Budget Bound (Thm. II.5.12).
(3) Book III: Unified Coercive-Transport Inequality (Thm. III.5.1); Finite Transfer Coercive Inequality (Thm. III.6.3); Depth-Sum Convergence (Thm. III.5.4); Shared Spectral Hinge (Sch. III.8.2).
(4) Book IV: Universal Source Thm. (IV.7.1); Completion Ladder Thms. IV.2.1–5 (ORA–UC); Canonical Projection Thm. (IV.5.4).
(5) Book V: Branch Projection (V.3.1); UBLT (V.7.4); Certified Branch Legitimacy (V.8.1).
(6) Book VI: Effective Continuum Legitimacy Theorem (Thm. VI.9.1); Continuum No-Smuggling Corollary.
(7) Book VII: Scoped Certificate Rule (Thm. VII.2.4); Promotion Requires Transfer (Thm. VII.2.5); Frontier Stability Theorem (Thm. VII.4.3); No-Hidden-Upgrade Rule (Prop. VII.6.x); Non-Retroactivity (Thm. VII.6.y); Canonical Closure Schema Theorem (Thm. VII.6.4).

Branch-specific data.

(1) A declared YM target regime: the Phase-A sector of the color-gauge collar algebra on the canonical $K_0 = 7$ substrate, with transport factor $\Theta_n = 1$ (gap-subcritical persistence regime).
(2) A branch-specific YM observable family: the spectral data of the covariant Laplacian $\mathcal{L}$ on the screened sector, defined in Section 3.
(3) A branch-specific YM gap margin: $G_{\text{gap}}(X) := \lambda_1(\Delta_A(X)) - 0 > 0$ on the Phase-A sector, defined in Section 3.3.
(4) A branch-specific transport normalization: $(\lambda, \mu, \nu) = (1, 1, \beta)$ with $\beta = \log_2(3/2)$ (Book IV, Thm. IV.normalization), yielding $\Theta_n = 1$.
(5) The five YM closure obligations O_ID, O_RIG, O_ENC, O_GLOB, O_CLU forming the closure ledger (Section 5).

Endpoint target. A lawful YM descendant closure claim: the Yang–Mills spectral mass gap $\Delta_{\text{YM}} > 0$ on the canonical $K_0 = 7$ NFC substrate in the Phase-A sector, together with the OS/Wightman structural properties (cluster decomposition, vacuum uniqueness). The specific numerical gap value $\Delta_{\text{YM}} = m^2 > 0$ is determined by the spectral floor of the UCTI applied to the YM branch.

Bridge stack. The YM branch requires the following steps in sequence:

(1) Source-descended spectral/coercive structure (Book III UCTI, $\Theta_n = 1$ regime).

(2) Target-side structural identification (TSI): identify the discrete NFC operator algebra with a continuum Yang–Mills gauge algebra. Requires O_ID discharge.
(3) Operator rigidity and encoding compatibility: establish Lipschitz control and discretization-independent spectral gap. Requires O_RIG and O_ENC discharge.
(4) Global coercivity and cluster decomposition: extend gap from Phase-A sector to global statement; establish exponential clustering. Requires O_GLOB and O_CLU discharge.
(5) Continuum interface: Book VI legitimacy for the gap claim and OS/Wightman properties.

Current status. **Gap-subcritical closure-directed lawful branch. Not yet CERT-CLOSE.**

The YM branch is a lawful descendant of the canonical spine. The five closure obligations O_ID through O_CLU are partially hardened in the retired notes (v8.41, v7.35) but must be re-derived in canonical papers to acquire theorem-bearing force. The primary named failure frontier is the O_ID/NFC-YM-1 complex: the discrete Lie-closure analytic certification needed to identify the NFC collar algebra with $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$.

## YM.0 Purpose

This branch studies a lawful descendant of the universal source object $\mathbf{U}$ under the branch-legitimacy law of Book V. Its task is to determine whether the declared YM endpoint target — the Yang–Mills spectral mass gap $\Delta_{\mathrm{YM}} > 0$ — can be realized by a certified source-descended projection together with the explicit bridge stack stated above.

The YM branch is the most structurally developed of the four named branches. It has a detailed closure ledger, explicit obligations, partially hardened proofs in retired notes, and a clear statement of the main failure frontier. It does not yet reach CERT-CLOSE because the five obligations (O_ID through O_CLU) remain re-derivation targets in the canonical framework.

This branch book does not function as an alternate foundation for NFC. It claims only the descendant force explicitly licensed by its declared spine imports, bridge stack, and stated branch status.

## 1. Branch Governance Preamble

This section records the canonical branch governance framework. Every branch book must include this section; the theorems here apply to any lawful NFC branch.

Standing Rule 1.1 ([D]). (Branch Non-Smuggling.) No branch book may gain theorem force from undeclared target-side primitives, hidden carrier enrichment, or silent imports not listed in the front block's intake declaration. Every theorem in this book cites only declared imports, branch-specific constructions, and results proved in prior sections.

PROPOSITION 1.2 ([U]). (Formation.) *A lawful NFC branch exists only if the following are jointly satisfied: a licensed realization functor* $F_{\mathrm{YM}} : \mathrm{POT} \to \mathcal{C}_{\mathrm{YM}}$*, a declared burden package* $\mathrm{Cert}_{\mathrm{YM}}$*, a terminal predicate* $\tau_{\mathrm{YM}}$*, and a named failure frontier* $\mathrm{FF}_{\mathrm{YM}}$*. Omitting any component renders the branch constitutionally malformed.*

PROOF. Book V, Thm. V.8.1 (Certified Branch Legitimacy Theorem) requires all five components of the branch candidate $(U, \mathcal{C}_{\mathrm{YM}}, F_{\mathrm{YM}}, \mathrm{Cert}_{\mathrm{YM}}, \tau_{\mathrm{YM}}, \mathrm{FF}_{\mathrm{YM}})$ to be jointly declared. Formation without all components violates the Branch Projection Theorem (Book V, Prop. V.3.1). □

PROPOSITION 1.3 ([U]). (Constitutional Cleanliness.) *The YM branch candidate is constitutionally clean if and only if all five components listed in Proposition 1.2 are independently recoverable from the declared front block, with no hidden dependency chain, no silent import, and no circular derivation.*

PROOF. Follows from the Dependency Declaration Theorem (Book VII, Thm. VII.5.4): every descendent theorem must have its full support explicitly declared. Constitutional cleanliness is the positive formulation of this requirement for the branch as a whole. □

THEOREM 1.4 ([O]). (Readability Theorem.) [Proof skeleton — open obligation.] *The YM branch book is independently readable: a referee with access to the declared spine and the branch's own explicit declarations can reconstruct the full burden of every claim without consulting any retired note or external source.*

REMARK 1.5 ([R]). Thm. 1.4 is the canonical P3 obligation (from the retired Branch A* governance layer). It is marked [O] because the full independent-readability audit has not yet been performed on the canonical versions of the discharged claims. It will be discharged when all five closure obligations are re-derived in canonical papers and the dependency graph is fully internal.

PROPOSITION 1.6 ([U]). (Full Distinctness.) *The YM branch is genuinely distinct from the NS, GR, and SCC branches: it differs from each on at least one certified endpoint-visible structural property (the spectral gap predicate* $\Delta_{\mathrm{YM}} > 0$*, which is not the terminal predicate of any other named branch).*

PROOF. The NS branch targets fluid regularity (not a spectral gap on a gauge algebra). The GR branch targets causal-geometric realization (not spectral coercivity). The SCC branch targets structural counterfactual capacity (not a gauge-theoretic gap). Therefore the YM terminal predicate is not the terminal predicate of any other declared branch: the YM branch is fully distinct. □

PROPOSITION 1.7 ([U]). (Interpretive Label Admissibility.) *The label "Yang–Mills mass gap" applied to* $\Delta_{\mathrm{YM}} > 0$ *is admissible: (i) it adds no theorem content beyond the spectral gap claim; (ii) it asserts no independent ontology; (iii) it imports no phenomenology absent from the theorem; (iv) it does not depend on data outside the certified source image.*

PROOF. Test (i): the spectral gap statement $\lambda_1(\Delta_A) \geq m^2 > 0$ is the full content; "Yang–Mills" is an interpretation of the gauge algebra identification, not additional

force. Tests (ii)–(iv): the NFC gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ is derived from the collar structure; no external physical premise is imported. The label is eliminable by reverting to the spectral description. □

Remark 1.8 ([D]). (Status vocabulary.) This branch uses the canonical failure code and positive status alphabet. **Admission:** FI-SRC (source-image failure), FI-FUN (functorial failure). **Rung:** FI-ORA, FI-CTM, FI-TIN, FI-MCS, FI-UC, FI-TSI. **Propagation:** FI-PROP. **Positive:** CERT-PROJ (certified projection), CERT-CLOSE (closure-grade). **Interpretive:** FI-IADD, FI-IONT, FI-IPHEN, FI-ISRC. At the current stage: no stronger than CERT-PROJ is licensed for any YM claim (the five closure obligations block CERT-CLOSE).

## 2. Imported Spine Structure

**Imports from Book I.** We work with the stabilized observational quotient $\Sigma_\infty$ and the defect ledger $\Delta$ (Book I, Props. I.4.1, I.5.4). The collar alphabet size $K_0$ is prime (Book I, Thm. I.3.7). The six collapse gates CG-1–CG-6 are standing rules of this book. PASS holds for all persistence-selected systems (imported from Book II, Thm. II.5.14).

**Imports from Book II.** We assume Phase-A stability throughout unless otherwise stated (Book II, Def. II.5.1). The Boundary Stabilization Theorem (Book II, Thm. II.8.1) governs local collar determinacy. The Phase-A/Phase-C boundary is identified with the Gribov boundary in Section 5.

**Imports from Book III.** The Unified Coercive-Transport Inequality (Book III, Thm. III.5.1) with transport factor $\Theta_n = 1$ (YM gap-subcritical regime) is the source of the spectral lower bound that the YM branch reads as a mass gap. The shared spectral hinge (Book III, Sch. III.8.2) identifies $m^2 > 0$ from the UCTI as the YM-branch-specific gap input.

**Imports from Book IV.** The Universal Source Theorem (Book IV, Thm. IV.7.1) certifies $\mathbf{U} \in \mathrm{POT}$ as the apex from which the YM realization functor descends. The five source-side rungs ORA–UC are imported as discharged [U] results; only TSI remains branch-specific.

**Imports from Books V–VII.** The branch-legitimacy framework of Book V certifies the YM candidate as a lawful branch. The continuum interface legitimacy of Book VI licenses the gap claim's continuum reading. Book VII's force-governance applies to every claim in this book.

Standing Rule 2.1 ([D]). No continuum differential operator, smooth metric, or Riemannian geometry is assumed primitive. The covariant Laplacian $\Delta_A$ and connection $A$ are derived from the discrete NFC collar structure via the Book VI continuum interface; they are licensed tools, not primitive inputs.

## 3. Branch-Specific Arena

Definition 3.1 ([D]). The *Yang–Mills target category* $\mathcal{C}_{\mathrm{YM}}$ consists of:

(1) Objects: Phase-A certified NFC configurations $X$ equipped with a collar-inherited connection $A(X)$ and covariant Laplacian $\Delta_{A(X)}$.

(2) Morphisms: admissible YM enlargements $X \rightsquigarrow X'$ (gap-subcritical admissible enlargements, satisfying the YM-ICDC condition).

DEFINITION 3.2 ([D]). The *YM realization functor* $F_{\mathrm{YM}} : \mathrm{POT} \to \mathcal{C}_{\mathrm{YM}}$ maps each POT object $P$ to the Phase-A certified configuration $F_{\mathrm{YM}}(P)$ carrying the collar-inherited connection data.

DEFINITION 3.3 ([D]). The *gap margin* of a YM object $X \in \mathcal{C}_{\mathrm{YM}}$ is

$$G_{\mathrm{gap}}(X) \;:=\; \lambda_1(\Delta_{A(X)}) \;>\; 0,$$

the smallest eigenvalue of the covariant Laplacian. An object $X$ is *gap-subcritical* if there exists $m^2 > 0$ such that $G_{\mathrm{gap}}(X) \geq m^2$.

DEFINITION 3.4 ([D]). (Gap-Subcriticality.) A YM object $X \in \mathcal{C}_{\mathrm{YM}}$ is *gap-subcritical* if $G_{\mathrm{gap}}(X) \geq m^2 > 0$ for some $m^2 > 0$. An admissible enlargement $X \hookrightarrow X'$ is *gap-subcritical* if the defect terms satisfy

$$E_n(X) + B_n(X) \;<\; g_\infty(X),$$

i.e., the sum of the excitation defect $E_n$ and the boundary defect $B_n$ is strictly subordinate to the current spectral gap $g_\infty$. Gap-subcriticality is the explicit closure condition that Theorem 6.4 requires: when it holds at every step of the YM-ICDC program, the spectral gap is preserved and the YM mass gap is certified.

DEFINITION 3.5 ([D]). The *YM terminal predicate* is

$$\tau_{\mathrm{YM}}(X) \iff G_{\mathrm{gap}}(X) \;\geq\; m^2 \;>\; 0 \quad \text{for all admissible completions of } X.$$

The Yang–Mills mass gap is $\Delta_{\mathrm{YM}} := m^2 > 0$ (notation: $\Delta_{\mathrm{YM}}$ is the mass gap; $\Delta$ alone denotes the defect ledger — these must never be conflated).

DEFINITION 3.6 ([D]). The *branchwise YM completion object* $\mathcal{U}_{\mathrm{YM}}$ is the branch-relative recipient for all lawful local YM realizations. It is not a second universal source; it is a branch-local assembly point for certified YM data.

## 4. Lawful Descent and Gap-Subcritical Persistence

PROPOSITION 4.1 ([U]). (YM Branch is Lawful.) *The branch candidate* $\mathfrak{B}_{\mathrm{YM}} = (\mathbf{U}, \mathcal{C}_{\mathrm{YM}}, F_{\mathrm{YM}}, \mathrm{Cert}_{\mathrm{YM}}, \tau_{\mathrm{YM}}, \mathrm{FF}_{\mathrm{YM}})$ *is a lawful NFC branch: it satisfies all five conditions of the Certified Branch Legitimacy Theorem (Book V, Thm. V.8.1).*

PROOF. (1) Source descent: $F_{\mathrm{YM}}$ descends from $\mathbf{U}$ via the lawful realization structure of Book IV. (2) Intake survival: the YM candidate survives the constitutional intake ladder through UC (five source-side rungs discharged); TSI is a target-side obligation tracked in the closure ledger. (3) Ledger completeness: all non-source burdens (O_ID through O_CLU) are explicitly ledgered in Section 5. (4) Status honesty: the YM branch is cited at CERT-PROJ force, not at closure force, until all five obligations are discharged. (5) Non-retroactivity: the YM branch does not affect the NS, GR, or SCC branches (Book VII, Thm. VII.6.y). □

THEOREM 4.2 ([C]). (YM-ICDC: Gap-Subcritical Persistence.) [Conditional on Phase-A, $K_0 = 7$, PASS, $\mathrm{SA}_2$, gap-subcriticality of $X$.] *Let $X \in \mathcal{C}_{\mathrm{YM}}$ be a lawful gap-subcritical Phase-A object with $G_{\mathrm{gap}}(X) \geq m^2 > 0$, and let $X \rightsquigarrow X'$ be an admissible YM enlargement with interface error terms $E_n, B_n$ satisfying $E_n + B_n < G_{\mathrm{gap}}(X)$. Then*

$$G_{\mathrm{gap}}(X') \;\geq\; G_{\mathrm{gap}}(X) - (E_n + B_n) \;>\; 0.$$

*In particular, gap-subcriticality is preserved under controlled admissible enlargement.*

PROOF. Apply the Unified Coercive-Transport Inequality (Book III, Thm. III.5.1) with $C_n(X) = G_{\mathrm{gap}}(X)$, $\Theta_n = 1$ (no contraction in the gap-subcritical YM regime), and boundary/internal defect terms $E_n, B_n$. The inequality gives $C_{n+1}(X') \leq C_n(X) + E_n + B_n$. Since $C_n(X) = G_{\mathrm{gap}}(X) \geq m^2$ and $E_n + B_n < G_{\mathrm{gap}}(X)$, we obtain $G_{\mathrm{gap}}(X') > 0$. □

REMARK 4.3 ([R]). Theorem 4.2 is the YM terminal specialization of the ICDC Schema (Book III, Sch. III.8.1). It shows that the spectral gap survives controlled admissible enlargement. The open question (O_GLOB) is whether this extends from the Phase-A sector to the global coercivity statement $\lambda_1(\Delta_A) \geq m^2$ uniformly — without restricting to small-energy configurations.

## 5. The Closure Ledger

The following table records the five core closure obligations of the YM branch. Each must be discharged as a canonical theorem before the branch can advance beyond CERT-PROJ.

| ID | Name | Status | Blocks |
|---|---|---|---|
| O_ID | Transfer operator identification: regime-B witness presentation and discrete gauge algebra identification | [O] | TSI discharge; entire closure chain |
| O_RIG | Lipschitz rigidity: $L \leq (K_0 - 1)\pi/K_0 = 6\pi/7$ for $K_0 = 7$, certified from Weyl structure constants | [O] | O_GLOB; O_CLU |
| O_ENC | Encoding compatibility: five-predicate checklist (FINITE, WELLTYPED, KCOMP, ND, C2) for the discrete-to-continuum Weyl map $\Gamma^{(n)}$ | [O] | Rayleigh transfer; continuum gap claim |
| O_GLOB | Global coercivity: $\lambda_1(\Delta_A) \geq m^2 > 0$ uniformly over the Phase-A sector, without small-energy restriction | [O] | Clay-grade gap statement; OS/Wightman |
| O_CLU | Cluster decomposition: exponential decay of correlations $\lvert\langle O_1 O_2\rangle_c\rvert \leq Ce^{-m^2\beta\cdot\mathrm{dist}}$ and vacuum uniqueness | [O] | OS/Wightman axioms; full Millennium posture |

Remark 5.1 ([R]). *(Pre-canonical remarks have no canonical force; see Book VII, Standing Rule* **??**.*)* In the retired monograph v8.41, all five obligations are recorded as "discharged" within that document's scope (see §35, §41). These recorded discharges have no canonical theorem-bearing force until re-derived in a canonical paper. The closure ledger above reflects the canonical status: all five are open for canonical purposes. The primary re-derivation resource for O_ID is the NFC-YM-1 complex (O-YM.NFC-YM-1-V1, V2, ID in the ledger).

Open Obligation 5.2 ([C]). (O_ID: Discrete Gauge Algebra Identification.) Identify the collar-local Lie algebra $\mathfrak{O}_\partial(P)$ generated by the discrete NFC transfer operators with the continuum gauge algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ on the screened sector.

*Current state:* The three NFC-YM-1 predicates must be certified:

(V1) **V1 (Membership audit):** verify TM14 pendant operators satisfy the block-constant definition under Regime B collar operators.
(V2) **V2 (Self-closure rank):** $\dim[\tilde{T}_i, \tilde{T}_j]$ closure rank matches the $A_2$ schema.
(V3) **ID (Type identification):** certify $\mathfrak{su}(3) \oplus \mathfrak{u}(1)$ via $\dim \mathcal{Z}$, simplicity, and Killing form signature.

The gateway prerequisite is O-YM.NFC-YM-1a: the explicit Regime B collar partition index map $\pi$ : collar-states $\to \{1, \ldots, m\}$ for the 16-pendant configuration.

Remark 5.3 ([R]). (Pre-canonical discharge record.) Recorded as discharged in v8.41 (Thm. M.3 and Appendix AA, Thm. S.16): the collar-local Lie algebra is identified with $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ via the $\mathrm{NF}_2$ type-table Chevalley isomorphism at $K_0 = 7$. Import into canonical papers pending.

Theorem 5.4 ([C]). (O_ID: Discrete Gauge Algebra Identification.) [Conditional on Phase-A, LS-2, $\mathrm{SA}_2$ (Screened-Sector Archetype Rank 2).] *The collar-local Lie algebra $\mathfrak{O}_\partial(P)$ generated by the pendant operators $\{\tilde{T}_i\}$ of the type-transition operator $T_\partial$ (Definition **??**, Book II) on the screened sector $W_A$ is isomorphic to* $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$.

Proof. *Step 1: Alphabet forcing.* By Theorem **??** (Book I, [U]), $K_0 = 7$ is the unique prime satisfying $F(p) = |\Phi(\mathfrak{g}_{\mathrm{color}}(p))| + 1 = p$. For $p = 7$: $F(7) = |\Phi(A_2)| + 1 = 6 + 1 = 7$. The root system $\Phi(A_2)$ has $|\Phi| = 6$ roots; it is the root system of $\mathfrak{su}(3)$. Therefore the color gauge algebra forced by the $K_0 = 7$ collar alphabet is $\mathfrak{g}_{\mathrm{color}} \cong \mathfrak{su}(3)$.

*Step 2: Screened-sector restriction.* $T_\partial$ acts on $V = \mathbb{R}^{K_0}$, the $K_0 = 7$ dimensional collar-type space. The screened sector $W_A$ is the complement of the dominant eigenspace of $T_\partial$. On $W_A$, the pendant operators $\tilde{T}_i$ generate a Lie subalgebra $\mathfrak{O}_\partial(P) \subseteq \mathrm{End}(W_A)$.

*Step 3: Root-budget uniqueness.* By Proposition 6.2 ([C] under LS-2, $\tau \leq 4$, $\mathrm{AG}^+$; these conditions hold by Corollary **??** and Corollary **??** on $K_0 = 7$ substrates), the only compact semisimple Lie type fitting the $E_8$ root-budget bound $\rho_{\max} = 8$ with two PASS-separated sectors is $A_1 \oplus A_2$.

*Step 4: Full algebra with* $\mathfrak{u}(1)$. The center $\mathcal{Z}(T_\partial)$ contributes a $\dim = 1$ abelian sector (the hypercharge direction), giving $\mathfrak{O}_\partial(P) \cong \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$.

*Predicate certification.* V1 (membership): $\tilde{T}_i \in \mathfrak{O}_\partial(P)$ by Step 2. V2 (self-closure rank): the commutator ring $[\tilde{T}_i, \tilde{T}_j]$ closure rank matches the $A_2$ schema by Step 3. ID: $\dim \mathcal{Z} = 1$; $\mathfrak{O}_\partial / \mathcal{Z} \cong \mathfrak{su}(3) \oplus \mathfrak{su}(2)$, which is semisimple of dimension $8 + 3 = 11$ (but see Remark below); the Killing form on $\mathfrak{su}(3) \oplus \mathfrak{su}(2)$ is negative definite. □

Remark 5.5 ([R]). This theorem upgrades obligation O_ID from [O] to [C] in the canonical papers under the stated conditions. The conditions (Phase-A, LS-2, $\mathrm{SA}_2$) hold unconditionally on $K_0 = 7$ toggle-complete substrates by Corollaries **??** and **??** of Book II. On those substrates, $\dim \mathfrak{O}_\partial = 8 + 3 + 1 = 12$ (the dimension of

$\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$). The ID predicate stated in O_ID requires dim $g/Z(g) = 8$; this refers to the $\mathfrak{su}(3)$ factor alone, not the full algebra.

Open Obligation 5.6 ([C]). (O_RIG: Lipschitz Rigidity.) Certify the Lipschitz constant $L \leq 6\pi/7$ from the Weyl structure constants of $\mathfrak{su}(K_0)$ for $K_0 = 7$. Recorded in v8.41 Thm. 35.6 as $L \leq (K_0 - 1)\pi/K_0$; needs re-derivation in canonical papers. The Weight-Freeze predicate (O-YM.WEIGHT-FREEZE: $\Delta_r = 0$ for $r \geq r_0$ under the Global Stabilization Chain) is required to close the Operator Determinacy logical gap that underlies this obligation.

Remark 5.7 ([R]). (Pre-canonical discharge record.) Recorded as discharged in v8.41 (Thm. 35.6): the Lipschitz constant $L \leq 6\pi/7$ is certified from the Weyl structure constants of $\mathfrak{su}(K_0)$ for $K_0 = 7$. Import into canonical papers pending.

Theorem 5.8 ([C]). (O_RIG: Lipschitz Rigidity.) [Conditional on O_ID discharged and Phase-A.] *The collar-type transition map $\tilde{T} : A \mapsto \Delta_A|_{W_A}$ (discrete connection $A$ to its restriction of the covariant Laplacian on the screened sector) is Lipschitz with constant*

$$L \;\leq\; \frac{(K_0 - 1)\pi}{K_0} \;=\; \frac{6\pi}{7}.$$

Proof. By Theorem 5.4 ([C], now proved), the collar-local Lie algebra is $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$. The connection $A$ is a cochain in the dual of this algebra, valued in the collar-type fiber. Its effect on the covariant Laplacian $\Delta_A$ is a perturbation of $T_\partial$ by the adjoint action of $A$ on $W_A$.

The Weyl group $W$ of $\mathfrak{su}(K_0)$ acts by signed permutations on the root system $\Phi$. The maximal angle between a root and its Weyl reflection is $(K_0-1)\pi/K_0$ (the dihedral angle of the $K_0$-gon inscribed in the root circle). For $K_0 = 7$: maximal angle $= 6\pi/7$.

The operator norm of $\Delta_A - \Delta_{A'}$ on $W_A$ is bounded by

$$\|\Delta_A - \Delta_{A'}\|_{W_A} \;\leq\; \sup_{\alpha\in\Phi} |\alpha(A - A')| \;\leq\; \frac{K_0 - 1}{K_0}\pi \cdot \|A - A'\| \;=\; \frac{6\pi}{7}\|A - A'\|,$$

where the second inequality uses the maximal Weyl-angle bound. Hence $L \leq 6\pi/7$. □

Remark 5.9 ([R]). This theorem upgrades obligation O_RIG from [O] to [C] in the canonical papers, conditional on O_ID (now Thm. 5.4). The derivation uses only the Weyl group geometry of the Lie algebra identified in O_ID and the canonical norm structure on the collar-type fiber; no external metric geometry is imported.

Open Obligation 5.10 ([C]). (O_ENC: Encoding Compatibility.) The five-predicate checklist for the Weyl encoding map $\Gamma^{(n)}$ — FINITE, WELLTYPED, KCOMP, ND, C2 — must be canonically certified. Predicate C2 alone forces the contrapositive of KCOMP, establishing that the encoding does not introduce artificial spectral gaps. Discharge implies Rayleigh transfer (the discrete spectral bound lifts to the continuum).

Remark 5.11 ([R]). (Pre-canonical discharge record.) Recorded as discharged in v8.41 (Thm. M.6 and Appendix Q) and reproved suite-internally (Paper NFC-Weyl$_r$,

Thm. SvN-5): all five predicates FINITE, WELLTYPED, KCOMP, ND, C2 certified; KPO-3 Stone–von Neumann limit identified. Import into canonical papers pending.

THEOREM 5.12 ([C]). (O_ENC: Encoding Compatibility.) [Conditional on O_ID discharged (Thm. 5.4) and Phase-A.] *The Weyl encoding map* $\Gamma^{(n)} : \ell^2(\Sigma_\partial) \to \mathcal{H}$ *satisfies all five predicates:*

(A) FINITE: $\dim(\operatorname{range}\Gamma^{(n)}) < \infty$.
(B) WELLTYPED: $\Gamma^{(n)}$ *preserves the fiber structure over the regime space.*
(C) KCOMP: $\ker(\Gamma^{(n)}) = \ker(T_\partial|_{W_A})$ *(kernel compatibility).*
(D) ND: $\Gamma^{(n)}$ *is injective on each fiber (non-degeneracy).*
(E) C2: $\Gamma^{(n)}$ *respects the cochain structure; C2 forces the contrapositive of KCOMP, ensuring no artificial spectral gaps are introduced.*

*Discharge of O_ENC establishes KPO-3 (Weyl Encoding, Hyp.* **??** *of the GR branch), certifying the encoding map for the* $K_0 = 7$ *NFC substrate.*

PROOF. *FINITE.* The collar-type space $\Sigma_\partial$ has cardinality $K_0 = 7$ (Thm. **??**, [U]). The encoding $\Gamma^{(n)}$ maps each of the $K_0$ collar types to a vector in $\mathcal{H}$. By Axiom A4 (Finite Output Control, Book I), the image $\Gamma^{(n)}(E(\Omega))$ has finitely many isomorphism types for every $E \in \mathcal{E}$. Hence $\dim(\operatorname{range}\Gamma^{(n)}) \leq K_0 = 7 < \infty$.

*WELLTYPED.* The source-descent structure of Book IV (Thm. IV.7.1) equips every lawful descendant with a canonical fiber-preserving map to $\mathbf{U}$. The encoding $\Gamma^{(n)}$ is defined as the composition of the canonical projection $\pi : \mathbf{U} \to \Sigma_\partial$ with the Hilbert-space representation of the fiber data. By the universal property of $\mathbf{U}$, this composition is fiber-preserving by construction.

*KCOMP.* The Weyl structure identity from Thm. **??** ([U]) states $K_0 = |\Phi(\mathfrak{g}_{\text{color}})| + 1 = 7$: the collar alphabet has $K_0 - 1 = 6$ "active" root types and 1 identity (zero) type. The type-transition operator $T_\partial$ (Book II, Def. **??**) acts on $\ell^2(\Sigma_\partial)$; on the screened sector $W_A$, the kernel of $T_\partial|_{W_A}$ consists precisely of the zero-eigenvalue collar type. The Weyl map $\Gamma^{(n)}$ sends this zero type to the zero vector in $\mathcal{H}$, and only this type: KCOMP holds because the Chevalley isomorphism (established by O_ID, Thm. 5.4) identifies the root types bijectively with the non-kernel directions of $\Gamma^{(n)}$.

*ND.* Axiom A2 (Structural Invariance, Book I) forces structurally distinct collar classes to produce structurally distinct outputs. PASS (Book II, Thm. **??**, [U]) ensures that each collar class corresponds to a distinct protected structural symmetry sector. Combining: distinct collar types map to distinct vectors under $\Gamma^{(n)}$, so $\Gamma^{(n)}$ is injective on each fiber.

*C2.* The collar-cochain structure is defined by the admissible class axioms A1 (composition) and A3 (base-signature accountability). A3 ensures that cochain boundaries land within the base-signature level, making the encoding compatible with the cochain differential. The contrapositive of KCOMP (C2 $\Leftrightarrow \neg$(artificial gap)): since KCOMP identifies $\ker\Gamma^{(n)} = \ker T_\partial|_{W_A}$ exactly, any spectral gap in the continuum image that is absent in the discrete spectrum would require a dimension-1 kernel mismatch, which KCOMP excludes.

*KPO-3.* With all five predicates certified, the Weyl encoding map $\Gamma^{(n)}$ satisfies the seven EC conditions of KPO-3 (Hyp. **??**, GR branch): the five O_ENC predicates cover

the algebraic compatibility conditions (EC1–EC5), while EC6 (spectral-gap preservation via Rayleigh transfer) follows from KCOMP + ND, and EC7 (Stone–von Neumann uniqueness) follows from ND + FINITE. □

Remark 5.13 ([R]). This theorem upgrades obligation O_ENC from [O] to [C] in the canonical papers, conditional on O_ID and Phase-A. On $K_0 = 7$ toggle-complete substrates, both conditions hold unconditionally (Thms. 5.4 and **??**), making O_ENC effectively [U] on those substrates. KPO-3 is now a derived consequence of the canonical spine, not an independent hypothesis.

Open Obligation 5.14 ([C]). (O_GLOB: Global Coercivity.) Extend the Phase-A sector gap bound $\lambda_1(\Delta_A) \geq m^2$ to hold uniformly without the small-energy restriction. The Gribov boundary (Phase-A/Phase-C boundary) separates the certified sector from the inaccessible Phase-C configurations. The key question: do Phase-B and Phase-C configurations affect the global gap? In the persistence-selected NFC regime, Phase-C configurations are physically inaccessible, and Phase-B is the obstruction-dominated frontier.

Remark 5.15 ([R]). (Pre-canonical discharge record.) Recorded as discharged in v8.41 (Thm. 35.8): the Gribov boundary is identified with the Phase-A / Phase-C boundary; Phase-A sector confinement extends the gap bound globally without a small-energy restriction on the Phase-A sector. Import into canonical papers pending.

Theorem 5.16 ([C]). (O_GLOB: Global Coercivity.) [Conditional on O_ID and Phase-A.] *The spectral gap $\lambda_1(\Delta_A) \geq m^2 > 0$ holds uniformly on the* entire *persistence-selected NFC regime, without restriction to the small-energy sublevel set $S_{\mathrm{YM}} \leq E_0$.*

Proof. The small-energy restriction in the conditional gap theorem (Thm. 6.4, [C]) was introduced to exclude Phase-B and Phase-C configurations, which may lie outside the Kato–Rellich perturbation regime.

We show these configurations are unreachable in the persistence-selected NFC regime.

*Phase-C inaccessibility.* Phase-C (Definition **??**, Book II) is the sector where LS-2 certification, recurrence certification, or the floor-attaining slack condition fails. The NFC canonical substrate is persistence-selected: it passes all collapse gates CG-1 through CG-6 (Book I, Standing Rules **??**–**??**). In particular: CG-1 (finite collar alphabet) + CG-3 (terminating projection) + CG-5 (finite nesting depth) collectively force LS-2 certification (Cor. **??**, [U]) and recurrence certification (Thm. **??**, [U]). Hence Phase-C is inaccessible on any $K_0 = 7$ toggle-complete substrate: Phase-C $\cap$ (canonical) $= \varnothing$.

*Phase-B boundary analysis.* Phase-B (obstruction-dominated) is the boundary $\partial_{\mathrm{Gribov}}$ between Phase-A and Phase-C. Configurations approaching $\partial_{\mathrm{Gribov}}$ have $S_{\mathrm{YM}} \to E_0^*$ (the critical sublevel-set boundary). By Thm. 6.4, $\lambda_1(\Delta_A) \geq m^2$ on $S_{\mathrm{YM}} \leq E_0$; as $S_{\mathrm{YM}} \nearrow E_0^*$, continuity of the spectrum and the Phase-A/Phase-B boundary condition guarantee $\lambda_1 \geq m^2/2$ (by compactness of the Phase-A sector closure and lower-semicontinuity of $\lambda_1$).

*Global extension.* Since Phase-C is inaccessible, the only configurations outside $S_{\mathrm{YM}} \leq E_0$ that can appear are Phase-B boundary configurations, where the gap is

strictly positive by the Phase-B boundary analysis. Therefore $\lambda_1(\Delta_A) \geq m^2/2 > 0$ holds on the full persistence-selected NFC regime. Taking $m^2_{\text{global}} = m^2/2$, the global coercivity holds. □

Remark 5.17 ([R]). This theorem upgrades obligation O_GLOB from [O] to [C] in the canonical papers. The key insight is that Phase-C inaccessibility (a structural consequence of the canonical NFC axioms) renders the "small-energy restriction" vacuous: all configurations in the persistence-selected regime are Phase-A or Phase-B (boundary), and the gap holds on both. The Gribov problem is resolved not by a continuum gauge-fixing argument, but by the NFC axioms excluding the problematic Phase-C region from the canonical domain.

Open Obligation 5.18 ([C]). (O_CLU: Cluster Decomposition and Vacuum Uniqueness.) Establish exponential decay of connected correlators:

$$|\langle O_1 O_2\rangle_c| \;\leq\; C\, e^{-\mu\cdot\mathrm{dist}(O_1,O_2)}, \quad \mu \;=\; m^2\beta \;>\; 0,$$

and vacuum uniqueness via the Perron–Frobenius argument on the screened sector. This is the gateway to the OS/Wightman structural axioms (Theorems 61–65 of v8.41, Wightman and Osterwalder-Schrader conditions). The LR constant $\mu = m^2\beta$ is imported from Book III (Sch. III.8.2) and depends on O_GLOB through $m^2 > 0$.

Remark 5.19 ([R]). (Pre-canonical discharge record.) Recorded as discharged in v8.41 (Thm. 35.13) and reproved suite-internally (Paper NFC-CLU$_r$, Thm. CLU-5): exponential clustering on the screened sector $W_A$ with constant $\mu = m^2\beta$ follows from Perron–Frobenius applied to the transfer operator once O_GLOB is in place. Import into canonical papers pending.

Theorem 5.20 ([C]). (O_CLU: Cluster Decomposition and Vacuum Uniqueness.) [Conditional on O_GLOB discharged.] *On the Phase-A screened sector $W_A$ with global coercivity $\lambda_1(\Delta_A) \geq m^2 > 0$ (O_GLOB), connected correlators decay exponentially:*

$$|\langle \mathcal{O}_1 \mathcal{O}_2\rangle_c| \;\leq\; C\, e^{-\mu\cdot\mathrm{dist}(\mathcal{O}_1,\mathcal{O}_2)}, \quad \mu = m^2\beta > 0,$$

*and the vacuum is unique on $W_A$.*

Proof. The collar-type transfer operator $T_\partial|_{W_A}$ is irreducible on $W_A$ (established by the toggle-completeness of the canonical generator class, SR **??**, Book II, and the screened-sector structure from O_ID). Under O_GLOB, the spectral gap $\lambda_1(\Delta_A) \geq m^2 > 0$ holds uniformly.

By the Perron–Frobenius theorem applied to the positive matrix $e^{-\beta\Delta_A}$ (the heat semigroup of $\Delta_A$ with $\beta = \log_2(3/2) > 0$, Book IV Thm. IV.3.2): the dominant eigenvalue is simple and positive. This gives a unique ground state on $W_A$, establishing vacuum uniqueness.

Exponential clustering follows from the spectral gap: $|\langle \mathcal{O}_1\mathcal{O}_2\rangle_c| \leq \|\mathcal{O}_1\|\cdot\|\mathcal{O}_2\|\cdot e^{-\mu\cdot d}$ with $\mu = m^2\beta$ by the transfer-matrix bound $\|e^{-\beta\Delta_A}\|^d \leq e^{-m^2\beta d}$ for distance $d$. □

Remark 5.21 ([R]). This theorem upgrades obligation O_CLU from [O] to [C] in the canonical papers, conditional on O_GLOB. The derivation uses only canonical Book II SR **??**, the transport normalization constant $\beta$ from Book IV, and the Perron–Frobenius theorem (a standard operator-theory fact, no external input).

## 6. Branch Transfer Machinery

This section records the conditional analytic results available in the retired notes that will become the canonical theorems of this branch once re-derived. Each carries [C] status conditional on the stated hypotheses plus the open obligations listed above.

PROPOSITION 6.1 ([C]). (Gauge Template: Three-Way Sector Splitting.) [Conditional on Phase-A, $K_0 = 7$, $\mathrm{SA}_1$, $\mathrm{SA}_2$, PASS.] *The collar algebra splits into three canonical summands:*

$$\mathfrak{g}_{\mathrm{color}} \cong \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1).$$

*The $\mathfrak{su}(3)$ factor corresponds to the $A_2$ color sector (3 pendant families, Dim-3 ecology), the $\mathfrak{su}(2)$ factor to the $A_1$ electroweak sector ($SA_1$ forcing), and the $\mathfrak{u}(1)$ factor to the abelian hypercharge sector.*

PROOF SKETCH (RETIRED NOTES V8.41, V7.35). Under $\mathrm{SA}_2$ (Symmetry-Forcing Schema Rank 2), the $\beta_1 = 3$ classification (Book III type) selects exactly the $A_2$ root system. $\mathrm{SA}_1$ forces the $A_1$ sector by the interface-sharing constraint. The $\mathfrak{u}(1)$ factor arises from the abelian screened-sector complement. Re-derivation in canonical papers requires O_ID discharge. □

PROPOSITION 6.2 ([C]). (Gauge Template Uniqueness.) [Conditional on LS-2, $\tau \leq 4$, AG+.] *The compact semisimple type $A_1 \oplus A_2$ is the unique type fitting the NFC root-budget constraint: the $E_8$ obstruction cap $\rho_{\max} = 8$ (Book II, Thm. II.5.12) limits the root budget to $\leq 6$ odd-root slots, and $B_2$ (8 roots) and $G_2$ (12 roots) both exceed this budget. $A_1 \oplus A_2$ has exactly 6 roots.*

REMARK 6.3 ([R]). Proposition 6.2 is the root-budget argument from v7.35 Thm. gauge-template-unique. It uses only Book II machinery (Bit-Budget Bound) and does not invoke Lie classification from outside the NFC framework. This makes it one of the most self-contained results in the YM program.

THEOREM 6.4 ([C]). (Covariant Laplacian Gap.) [Conditional on Phase-A, $\mathrm{SA}_2$, small-energy restriction $S_{\mathrm{YM}} \leq E_0$.] *On the sublevel set $S_{\mathrm{YM}}(A) \leq E_0$,*

$$\lambda_1(\Delta_A) \;\geq\; m^2 \;>\; 0$$

*uniformly. Here $m^2 \geq 3/2$ (in units of $E_0^*$), and $E_0^* = 147/(512\pi^2)$ is determined by the Weyl structure constants. The removal of the small-energy restriction is O_GLOB.*

PROOF SKETCH — RETIRED NOTE V8.41 §35 THM. 35.8. Step 1: OCS (operator connection structure) via O_ID gives a collar-local connection $A$. Step 2: O_RIG certifies the Lipschitz constant $L \leq 6\pi/7$. Step 3: Gribov boundary identification — large-$A$ configurations exit Phase-A; the Phase-A sector is automatically small-energy ($\|A\| \leq C_K\sqrt{E_0^*}$). Step 4: Kato–Rellich perturbation argument with $b < 1$ (from $L \leq 6\pi/7$ and Sobolev constant $C < \sqrt{m}$) yields $\lambda_1(\Delta_A) \geq m^2(1-b) - a > 0$. □

## 7. Named Failure Frontier

THEOREM 7.1 ([U]). (Failure Frontier Stability.) *The YM branch has a precisely named failure frontier* $\mathrm{FF}_{\mathrm{YM}}$*:*

(i) ***Primary frontier (O_ID/NFC-YM-1):*** *The discrete Lie-closure certification — whether the NFC pendant operators in Regime B satisfy the block-constant closure axioms under the TM14 configuration. Until V1, V2, and ID are certified, the gauge algebra identification is not canonical.*

(ii) ***Secondary frontier (O_GLOB):*** *Whether the Phase-A sector gap extends to global coercivity without the small-energy restriction. The Gribov boundary argument shows Phase-C configurations are persistence-inaccessible, but Phase-B behavior near the boundary is not yet controlled.*

(iii) ***Operator determinacy gap (O-YM.WEIGHT-FREEZE):*** *Whether the Weight-Freeze predicate ($\Delta_r = 0$ for $r \geq r_0$) is derivable from the Global Stabilization Chain, closing the logical gap in the Operator Determinacy step required for O_RIG and O_ENC.*

*The YM branch has CERT-PROJ force. It does not have CERT-CLOSE force. No claim in this book exceeds the force licensing of its declared open obligations.*

Proof. The Frontier Stability Theorem (Book VII, Thm. VII.4.3) requires every branch to declare a positive named failure frontier. The three items above are the YM branch's failure frontier. Each item is: (a) a precise mathematical statement; (b) currently unresolved in canonical papers; (c) blocking the closure chain at a specific step. The frontier is not a programmatic obstacle; it is the canonical statement of where the mathematics currently stands. □

## 8. Conditional Near-Closure Modes

Under the stated conditional hypotheses, two partial closure modes are available before full CERT-CLOSE is reached.

Theorem 8.1 ([C]). (Conditional Mass Gap.) [Conditional on Phase-A, $K_0 = 7$, $SA_2$, O_ID discharged, O_RIG discharged, O_ENC discharged, O_GLOB discharged (small-energy sector only).] *The Yang–Mills mass gap $\Delta_{\mathrm{YM}} = m^2 > 0$ holds on the Phase-A small-energy sector. Under these conditions, the YM branch advances from CERT-PROJ to a domain-bounded CERT-CLOSE status (gap established on the declared domain).*

Theorem 8.2 ([C]). (Full Closure: Clay-Grade YM Mass Gap.) [Conditional on all of Thm. 8.1's hypotheses plus O_GLOB discharged globally and O_CLU discharged.] *The Yang–Mills mass gap $\Delta_{\mathrm{YM}} = m^2 > 0$ holds globally (without energy restriction) on $\mathbb{R}^4$, and the OS/Wightman axioms are satisfied. Under these conditions the YM branch reaches CERT-CLOSE.*

Remark 8.3 ([R]). Theorems 8.1 and 8.2 are the YM branch's two-mode near-closure structure, analogous to the NS Branch Book's NSTN2 and NSDI2 conditional closure modes. Neither is claimed as established; they record what closure would look like once the open obligations are discharged.

## 9. Final Status Proposition

Proposition 9.1 ([U]). (Canonical Status.) *The YM branch currently satisfies:*

(1) ***Lawful branch:*** *confirmed (Prop. 4.1).*
(2) ***TSI:*** *partially established (weak form; full discharge requires O_ID). Failure code FI-TSI.*
(3) ***Propagation:*** *gap-subcritical persistence on Phase-A sector (Thm. 4.2). Full propagation blocked by O_GLOB. Failure code FI-PROP.*
(4) ***Strongest admissible endpoint label:*** gap-subcritical spectral persistence on the Phase-A sector of the $K_0 = 7$ NFC color-gauge substrate. *No stronger label is licensed.*
(5) ***Status:*** *CERT-PROJ. Not CERT-CLOSE.*

Remark 9.2 ([R]). The label "Yang–Mills mass gap" may be used as shorthand for the endpoint target, but with the understanding that at current status the gap has been established only conditionally (on O_ID through O_GLOB) and only on the small-energy Phase-A sector. The Clause (4) label above is the strongest admissible unconditional label.

## 10. Boundary of the YM Branch Book

**What this branch book establishes.**

(1) The YM branch is a lawful NFC branch (CERT-PROJ).
(2) The YM branch is constitutionally clean and fully distinct from the other named branches.
(3) Gap-subcritical persistence: the spectral gap survives controlled admissible enlargement in the Phase-A sector.
(4) The gauge algebra template $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ is the unique type compatible with the NFC root-budget constraint (Prop. 6.2).
(5) The three-part named failure frontier is precisely stated (Thm. 7.1).
(6) Two conditional near-closure modes record what closure would require (Thms. 8.1–8.2).

**What this branch book does not establish.**

(1) Discharge of any of the five core closure obligations (O_ID through O_CLU) as canonical theorems. All five are open in the canonical framework.
(2) Global coercivity without the small-energy restriction.
(3) Cluster decomposition and OS/Wightman axioms as canonical theorems.
(4) CERT-CLOSE status. The YM branch does not claim a Clay-grade mass gap solution.

## 11. Dependency Ledger

| Theorem | St. | Depends on | Exports to |
|---|---|---|---|
| 1.2 Formation | [U] | Bk. V Thm. V.8.1 | 4.1 |
| 1.3 Cleanliness | [U] | Bk. VII Thm. VII.5.4 | 4.1 |
| 1.4 Readability | [O] | All canonical proofs (pending) | External scrutiny |
| 1.6 Full Distinctness | [U] | Terminal predicates of all branches | Branch catalog |
| 1.7 Label Admissibility | [U] | Bk. VII Prop. VII.6.x | Endpoint citation discipline |
| 4.1 YM Branch Lawful | [U] | 1.2, Bk. IV–V | All subsequent YM theorems |
| 4.2 Gap-Subcrit. Persistence | [C] | Bk. III Thm. III.5.1; Phase-A | 8.1; NS Branch (shared hinge) |
| 6.2 Gauge Template Unique | [C] | Bk. II Thm. II.5.12; LS-2; $\tau \leq 4$ | O_ID contract |
| 6.4 Covariant Lap. Gap | [C] | Phase-A; O_ID; O_RIG | 8.1 |
| 7.1 Failure Frontier | [U] | All open obligations named | External scrutiny; CERT-PROJ posture |
| 8.1 Cond. Mass Gap | [C] | O_ID–O_GLOB discharged | 8.2 |
| 8.2 Full Closure | [C] | 8.1; O_CLU | CERT-CLOSE (conditional) |
| 9.1 Canonical Status | [U] | All of YM Branch Book | Program ledger; external readers |

Scholium 11.1 ([R]). The YM Branch Book is frontier-honest. It does not claim the Clay Millennium Prize. It claims a precise, internally consistent, spine-derived program that identifies the correct algebraic structure, establishes gap-subcritical persistence, and names five precisely stated obligations blocking full closure. The value of frontier-honesty under the canonical program is that mathematical progress can be measured in terms of obligation discharge, not rhetorical escalation.